MILITARY SPECIFICATION

ICEMAKING PLANT: BLOCK

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an electric-motor-driven, ammonia, ice-making plant for making 300-pound blocks of ice.

* 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-C-105	-	Calcium Chloride, Dihydrate and Calcium Chloride, Anhydrous, Technical
W-C-375	-	Circuit Breaker, Molded Case, Branch-Circuit and Service
W-P-115	-	Panel, Power Distribution
CC-M-641	-	Motor, Alternating Current, (Integral Horsepower 200 HP and Smaller)
HH-I-551	-	Insulation Block, Pipe Covering and Boards, Thermal (Cellular Glass)
HH-I-557	-	Insulation, Pipe, Valve and Fitting Covering, Thermal, Cork
HH-I-558	-	Insulation, Blocks, Boards, Blankets, Felts, Sleeving (Pipe and Tube Covering) and Pipe Fitting Covering, Thermal (Mineral Fiber Industrial Type)
QQ-S-775	-	Steel Sheets, Carbon, Zinc-Coated
QQ-S-781	-	Steel, Strapping, Flat
WW-H-171	-	Hangers and Supports, Pipe
GGG-P-323	-	Ice, Pick

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PPP-B-601 - Box, Wood, Cleated-Plywood
PPP-B-621 - Box, Wood, Nailed and Lock Corner
PPP-B-636 - Box, Fiberboard
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-V-173 - Varnish, Moisture-and-Fungus Resistant, (for the Treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank
MIL-T-704 - Treatment and Painting of Materiel
MIL-F-2312 - Felt, Cattlehair or Wool: Mildew Resistant, and Moisture Resistant, Treated
MIL-R-3593 - Refrigeration and Cooling Equipment (Excluding Household Refrigerators) Packaging of
MIL-F-3541 - Fittings, Lubrication
MIL-G-3859 - Grease Guns, Hand, High Pressure, Lever-Operated; Cartridge and Bulk Loading (14 ounce and 21 ounce Capacity)
MIL-G-10924 - Grease, Automotive and Artillery
MIL-B-11040 - Belt, V; Engine Accessory Drive
MIL-T-12625 - Thermometer, Self-Indicating, Liquid in Glass, Pocket Type
MIL-T-12664 - Treatment, Fungus Resistant, Paranitrophenol, for Cork Products
MIL-P-14105 - Paint, Heat-Resisting, (for Steel Surfaces)
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw
MIL-M-46032 - Mildew-Resistant Treatment (for Fabrics) Copper Processes.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U. S. Military Property
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics Measurement of
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods

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MIL-STD-27900 - Gage, Compound, Pressure and Vacuum Dial Indicating, Panel Mounted, Refrigerant-12
MIL-STD-27901 - Gage, Pressure, Dial Indicating, Panel Mounted, Refrigerant-12

DRAWING

Office of Chief of Engineers
T043-23, Storage Facilities Ice Plant, 15-Ton
40 x 60 Standard Frame

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

Standard IC-1-1965 - Industrial Control

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, N. Y. 10017.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL Standard 207C Refrigerant - Containing Components

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Ill. 60611; 1285 Walt Whitman Rd., Melville, Long Island, New York 11746; or 1655 Scott Blvd., Santa Clara, California 95050.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code

(Application for copies should be addressed to the American Society for Mechanical Engineers 345 E. 47th Street, New York, N. Y. 10017.)

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AIR-CONDITIONING AND REFRIGERATION INSTITUTE (ARI)

Standard 511 - Ammonia Compressors and Compressor Units

(Application for copies should be addressed to Air-Conditioning and Refrigeration Institute, 1346 Connecticut Avenue, N. W., Washington, D. C. 20006.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A153 - Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

A269-65- Seamless and Welded Austenitic Stainless Steel Tubing for General Service

D1621-59T - Test for Compression Strength of Rigid Cellular Plastics

D2016-65 - Methods of Test for Moisture Content of Wood

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

BUREAU OF MINES

List of Respiratory Protective Devices

(Application for copies should be addressed to the Bureau of Mines, Branch of Publications, 4800 Forbes Ave., Pittsburgh, Pa. 15213.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Ill. 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N. W. Washington, D. C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

* 3.1 First Article.- This specification contains provisions for First Article test and approval (see 4.2, 6.2, and 6.4).

* 3.2 Standard component.- The ice making plant shall be the manufacturer's current standard product except for any changes necessary to comply with the specification requirements.

* 3.3 Safety standards.- All materials and electrical equipment shall conform to UL standards for safety, as applicable.

* 3.3.1 Compliance.- Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment of ice making plants, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the materials and electrical equipment he proposes to furnish under this specification meets the requirements of the applicable UL standards. Acceptable evidence of meeting the requirements of these standards shall be the UL label or listing mark, or a certified test report, acceptable to the Government from a recognized independent testing laboratory, indicating that the materials and electrical equipment have been tested and conform to the requirements of the applicable UL standards.

* 3.4 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer for ice making plants, provided the completed item complies with all provisions of this specification. Commercial tolerances shall apply to all metal thickness and diameters unless otherwise specified herein.

* 3.4.1 Dissimilar metals.- Dissimilar metals in contact with each other shall be avoided unless protected against galvanic corrosion. Dissimilar metals shall be isolated from one another in accordance with MIL-T-704.

* 3.4.2 Fasteners.- All steel fasteners shall be zinc or cadmium plated. Sheet metal or self-tapping screws shall not be used in an area for structural strength. Threads shall conform to handbook H28.

* 3.4.3 Corrosion-resisting steel.- Corrosion-resisting steel shall conform to grade TP 304L of ASTM A269.

* 3.4.4 Pipe insulation.- Pipe insulation shall conform to HH-I-558, HH-I-557 or HH-I-551.

* 3.5 Design and construction.- The icemaking plant shall be furnished unassembled and shall consist of the following components: Compressor, condenser, receiver, evaporator, accumulator, oil separator, brine tank with agitator, ice cans, motors, harvesting equipment, necessary piping with shut-off valves for connecting components, calcium chloride for brine,

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ammonia refrigerant, all insulation materials and other accessories as specified herein. The icemaking plant shall include all equipment, exclusive of a source of electrical power, necessary for the manufacture of opaque ice from potable water. Template and anchor bolts for placing the compressors and motors on mounting bases shall be furnished. Design of the icemaking plant shall permit assembly of the plant in a building conforming to Drawing TO-43-23, referenced in 6.3.1. The icemaking plant shall operate manually except for automatic safety controls for plant components. Covers or plates for component adjustment repair, replacement, or maintenance shall be provided. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage. The icemaking plant shall produce not less than 15 tons of ice per 24 hours when operating in an ambient air temperature of 93 degrees Fahrenheit (F) dry bulb and 83 F. wet bulb with water supplied to the ice cans at a temperature of not less than 90 F. and a brine temperature of not more than 15 F. When the icemaking plant is operating at the conditions specified herein, the condensing pressure shall be not more than 190 pounds per square inch gage (psig) and the refrigerant suction pressure at the tank accumulator not less than 19 psig (see 4.4.3).

* 3.5.1 Compressor.- The compressor shall be a water-cooled, single acting, multiple V-belt reciprocating type, and shall be in accordance with the ARI Standard 511. Unless otherwise specified see 6.2, the compressor shall be equipped with isolation pads or springs and cross-over connections for unloading. The lubrication system shall be a forced-feed type to furnish lubrication to the compressor when operating in either direction of rotation. The compressor shall be driven by an electric motor as specified in 3.5.8. V-belts shall conform to MIL-B-11040. The recommended standard equipment shown in ARI Standard 511 and in addition the necessary oil charging hoses shall be furnished.

* 3.5.2 Condenser.- The condenser shall be an evaporative type and shall be equipped with a chemical water treatment device, bleed valve, and blow-off valve to prevent the buildup of slime, scale, and solids in the condensing water. The fan housing shall be constructed of galvanized sheet steel. The water tank section shall be constructed of zinc-coated sheet steel having a minimum thickness of 14 gage (.0785 inch nominal) for the bottom and a minimum thickness of 16 gage (.0635 inch nominal) for the sides. The condenser shall be equipped with a water circulating pump sized to furnish water for the condenser and a supply of cooling water to the refrigeration compressor. Circulating water pump for the condenser shall be of the centrifugal type, and the fans, fan scrolls and the primary pipe condensing surface shall be hot-dip zinc-coated in accordance with ASTM A153.

* 3.5.3 Receiver.- The liquid ammonia receiver shall hold not less than the complete refrigerant charge in 80 percent of its volume. The receiver shall be equipped with mounting supports and all necessary valves and fittings.

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Fittings and valves shall include, but not be limited to, dual safety relief valve; liquid level sight glass; and purging, charging, drain, inlet, and outlet valves. The receiver shall be in compliance with the ASME Boiler and Pressure Code.

3.5.4 Evaporator.- The surface of the evaporator coil and its arrangement shall be designed to produce ice in the capacity and conditions as specified in 3.5. The evaporator coil shall be equipped with an oil blow-off valve. Refrigerant feed to the evaporator coil shall be controlled by a low-pressure float-control valve equipped with valves, strainer, equalizer connection and a bypass line equipped with a manually controlled expansion valve.

* 3.5.5 Accumulator.- An accumulator shall be furnished with the evaporator coil to provide refrigerant flooding to the evaporator and to prevent liquid refrigerant from returning to the compressor. The accumulator shall be in compliance with the ASME Boiler and Pressure Code for accumulators.

* 3.5.6 Oil separator.- A compressor discharge line oil separator complete with flanged connections, drain valve, and lugs for overhead mounting shall be provided. The oil separator shall be equipped with an external float return that will automatically return the oil to the compressor crankcase. The oil separator shall be designed, fabricated, tested, inspected, and stamped in compliance with the ASME Boiler and Pressure Detail Code.

* 3.5.7 Brine tank.- The brine tank shall be of sheet steel not less than 3 gage (0.2391 inch nominal) thickness. The tank shall be painted as specified in 3.12 after fabrication and assembly. The tank shall be shipped knocked-down for assembly by welding, and shall be complete with internal braces, baffles, bulkheads, and trunk or raceway to the evaporator coil. Steel plates are to be furnished beveled where butt welds are required for assembly.

* 3.5.7.1 Agitator.- The agitator for circulating the brine through the raceway and throughout the brine tank shall be a vertical-shaft propeller-type driven by an electric motor. The agitator shall produce sufficient velocity of brine through the can space to insure uniform and adequate cooling to all cans when tested as specified in 4.4.3.

3.5.7.2 Thermometer.- The brine tank shall be equipped with a 45 degree angle thermometer having not less than a 5-inch scale, a range of minus 10 F. to plus 110 F., and an extended-type insulated stem.

3.5.7.3 Ice cans.- The ice cans shall be constructed of zinc-coated steel having a minimum thickness of 16 gage (0.0635 inch nominal). All seams shall be welded watertight, coated with heavy density zinc base coating to replace zinc-coating damaged in the welding operation and then smoothed.

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The cans shall be tapered to permit nesting inside each other for convenience in transportation and storage. Each can shall have a reinforcing band around the top edge. The cans shall be grooved. Each icemaking plant shall be equipped with 200 ice cans. Each can shall have a capacity of 300 pounds of ice. The top inside dimensions shall be 11 by 22 inches, and 51 inches deep.

3.5.7.3.1 Can clips.- Galvanized hold-down clips, wooden rests for cans, and black steel guides to retain cans in the correct position with relation to the brine level in the tank shall be provided.

* 3.5.7.4 Frames and covers.- Frames and covers shall be made of white oak, red oak, or other species of lumber with equal properties. The lumber shall be free of defects other than small surface checks, slight cross-grain, and pan knots, and shall be dried to a moisture content not to exceed 20 percent (see 4.4.6). Can covers and brine tank raceway covers shall be sealed against absorption of moisture by being hot dipped in linseed oil prior to final finishing. The covers shall consist of two layers of lumber, each layer consisting of 13/16-inch (finished) stock set at right angles to each other. The layers shall be nailed with drive screw spiral-type zinc-coated nails. The covers shall be smooth to facilitate cleaning. Each can cover shall be of sufficient size to cover two ice cans. A minimum of 90 covers shall be furnished with each icemaking plant. The ledger plate with expansion bolts which support the tank covers along the concrete wall of the brine tank inclosure shall be furnished as part of the cover and frame assembly.

* 3.5.8 Electric motors.- Electric motors shall conform to CC-M-641, and shall be continuous duty, style A-3, and fungusproof. Motors shall operate from a 208-volt, 60-hertz (Hz), 3-phase, unless otherwise specified (see 6.2). The rating of the motors shall be not less than the horsepower requirements of the driven equipment. The motors or starters shall be equipped with thermal overload protectors sized to prevent motor operation above the safe operating temperature for which the motor is rated. An arrow indicating the direction of rotation of the motor shall be provided on all motor frames. The compressor drive motor shall be not less than 60 horsepower (HP). Sixty Hz speeds must be maintained in order to secure required capacities. Equipment shall be provided with all auxiliary or additional parts, pulleys, belts and gears necessary to secure a conversion from operation on 60 Hz frequencies to 50 Hz. Motors and controllers shall be readily convertible in the field from 60 to 50 Hz operation and vice versa. All thermal and trip elements, relays, coils, springs and tools, required to accomplish the conversion shall be packed in a suitable kit and properly identified. Complete instructions shall be provided outlining the sequence and steps to be taken for conversion. Equipment having direct connected motors that cannot be converted to belt or gear drive shall be sized to give the required output at 50 Hz and be capable of being operated at increased capacities at 60 Hz.

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3.5.8.1 Controllers.- Each motor shall be provided with an enclosed controller conforming to NEMA IC 1. Control equipment shall be rated for the maximum voltage and current conditions encountered.

* 3.5.9 Electric-distribution panel.- The electric-distribution panel shall conform to class 1, W-P-115. The panel shall be equipped with automatic circuit breakers conforming to W-C-375 for both main and branch circuits, for individual motors, and for lighting. Electrical installation shall conform to the National Electrical ode.

3.5.10 Safety controls.- Safety controls shall be provided and shall consist of adjustable high-pressure and low-pressure refrigerant switches of the manual reset type, oil pressure failure switch, and a 4-inch alarm bell with transformer for mounting on the control panel, and connected so it will ring when the compressor stops due to high condensing pressure, low suction pressure, or loss of oil pressure.

3.5.11 Harvesting equipment.- Harvesting equipment shall consist of an electric hoist, a can dump, can-filling equipment, and ice chute.

* 3.5.11.1 Electric hoist.- The electric hoist shall be of the overhead type with a rated capacity of not less than 1/2 ton when operating at a hoisting speed of not less than 10 ft. per minute. An overhead track hoist conveying system shall be provided for harvesting continuously and simultaneously two 300 pound blocks of ice. The hoisting motor shall be provided with a manual starter, limit switches, electric trolley conductors, and sufficient wiring to connect with the electric-distribution panel.

* 3.5.11.2 Can dump.- A two 300 pound ice block capacity, manually operated can dump constructed of steel framework lined with wood to protect the cans shall be provided. The can-dump shall be equipped with an automatic water spray to thaw the ice loose from the cans. The can dump shall function to convey the ice to the storage room via the ice chute.

3.5.11.3 Can-filling equipment.- The can-filling equipment shall be an automatic pressure type can-filler.

3.5.11.4 Ice chute.- The ice chute shall be suitable for installation in the adjoining wall between the icemaking plant and the ice storage warehouse. The chute shall consist of a reinforced wall frame with a counterweighted swinging flap and inclined ramp. The chute shall be constructed of zinc-coated steel conforming to type I of QQ-S-775 with weight of metal suitable for the purpose. The counterweighted flap shall be hinged at the top, insulated and sealed against the passage of air by means of felt strips on the sides and a canvas strip along the lower edge. The felt stripping shall conform to type III of MIL-F-2312. The canvas stripping shall be treated for mildew resistance in accordance with type I of MIL-M-46032, grade A. The chute shall accomodate the passage of the size and number of ice blocks handled by the can dump.

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3.5.12 Refrigerant pipe.- All valves, pipe, and fittings required for the refrigerant pipe system shall be provided. Pipe shall be black steel and free of inside scale. Unassembled pipe ends shall be capped or plugged. All high side piping smaller than 2 inches in diameter shall be schedule 80, and all low side piping shall be schedule 40. Fittings shall be either the welding type or flanged type, except that fittings 3/4-inch or less may be screw type.

3.5.13 Water piping.- All valves, pipe, and fittings required for the complete water supply and drain piping shall be provided including connections to can-filler tank, can dump, brine tank, compressor water jackets, and evaporative condenser. All necessary control valves for servicing and maintenance shall be provided. Waterpiping shall be galvanized steel and unassembled pipe ends shall be capped or plugged.

* 3.5.14 Insulation.- Molded, lag, or built-up type pipe insulation shall be provided to cover all refrigerant suction lines, low-temperature liquid lines float-control valve, and the accumulator. Pipe insulation shall conform to HH-I-558, or the thermally equivalent thickness of vegetable cork conforming to HH-I-557, or cellular glass conforming to HH-I-551. Insulation shall also be provided to cover the sides, ends, and floor of the brine tank as follows: The floor of the brine tank shall be insulated with 4 inches of closed cellular non-water absorbent board insulation having a minimum commercially acceptable "K" factor of 0.28 BTU per square foot per inch thickness with a maximum compression set of 0.1 inch when loaded with a static load of 6 psi and applied in at least two layers with staggered joints. The sides and ends of the brine tank shall be insulated with 10 inches of commercially rated granular insulation; or a combination of granular and closed cellular, non-water absorbent insulation sheet board applied in layers with staggered joints and providing a maximum "U" factor of 0.05 BTU per square foot per hour per degree Fahrenheit. Insulation shall also be provided to cover the floor, walls, and ceiling of the ice-storage room. Ice-storage room insulation shall be of the same type and thickness as specified for the floor of the brine tank. Asphalt or other cementing material recommended by the manufacturer of the insulation shall be provided for the application of the board-type insulating material. All insulating material shall be rendered fungus resistant in accordance with MIL-T-12664 and all insulation shall be resistant to the absorption of moisture.

3.6 Accessories.-

* 3.6.1 Cooling coils.- Schedule 40 black pipe cooling coils for the ice-storage room shall be provided to maintain the temperature of the ice-storage room at between 24 F. and 28 F. The cooling coils shall be complete with necessary pipe hangers, rods, eye nuts, brackets, saddles, yard pipe stanchions, for piping to and from evaporative condenser, and other refrigerant and water piping, coil supports solenoid valves, expansion valves, and necessary hardware for mounting the coils to the ceiling of the ice-storage room. The coils shall be equipped with hot-gas connections for defrosting. The hangers shall conform to WW-H-171.

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* 3.6.2 Room thermostat.- One room thermostat with a range -10 to +50 F. to be used in conjunction with the ice-storage room cooling coils shall be provided.

* 3.6.3 Calcium chloride.- An initial charge of calcium chloride conforming to O-C-105 shall be provided for the preparation of the brine. The initial charge shall be sufficient to provide a brine freezing temperature not to exceed minus 10 F. Besides the initial charge, an additional 2,000 pounds of calcium chloride shall be furnished with each plant.

3.6.4 Salt scale.- A salt scale salinometer, graduated in degrees, for measuring brine density shall be provided with each ice plant.

3.6.5 Ice tongs.- Two pairs of commercial type ice tongs for handling the ice blocks shall be provided with each icemaking plant.

* 3.6.6 Pocket-type thermometer.- Two thermometers conforming to MIL-T-12625 shall be provided with each icemaking plant.

3.6.7 Refrigerant.- Two complete charges of refrigerant 717 (ammonia, NH_3) shall be provided with each icemaking plant. Fifty percent of the refrigerant shall be contained in returnable-type cylinders to insure sufficient heavy cylinders for pump-down storage while making repairs to the system. The remaining refrigerant shall be contained in Interstate Commerce Commission approved nonreturnable cylinders.

* 3.6.8 Gages.- High-pressure and compound pressure gages conforming to MS27901 and 27900 for measuring the compressor discharge and suction pressure shall be provided on a gage board.

3.6.9 Welding rod.- Sufficient welding rods in the types required to assemble the plant shall be provided with each icemaking plant.

3.6.10 Platform scale.- A beam-weighted platform scale for weighing the refrigerant cylinders shall be provided with each icemaking plant.

3.6.11 Masks.- Two ammonia type gas masks (U. S. Bureau of Mines, type "C" Cannister) and six extra cannisters shall be provided with each icemaking plant.

3.6.12 Oil-charging pump.- An oil-charging pump for adding oil to the compressor shall be provided with each icemaking plant.

* 3.6.13 Miscellaneous operating accessories.- The following supplies shall be provided with each ice making plant:

(a) Two (2) dozen ice picks conforming to GGG-P-323.
(b) One (1) Ph brine testing outfit.
(c) Two (2) dozen rolls litmus paper.
(d) One (1) gross 6-inch sulphur tapers.

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3.7 Electromagnetic interference.- The electromagnetic interference characteristics of the icemaking plant shall conform to class IIB of MIL-STD-461 (see 4.4.4).

3.8 Lubrication.- All surfaces and points requiring lubrication shall be provided with a means for lubricating and access.

3.8.1 Lubricants.- The icemaking equipment shall operate as specified herein when lubricated with military lubricants.

* 3.8.2 Lubrication fittings.- Lubrication fittings shall conform to type I of MIL-F-3541. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts.

* 3.8.3 Grease lubrication.- Grease lubrication shall be with the following:

a. MIL-G-23827 All sealed bearings and other bearings when operating temperatures do not exceed 250 F.

b. MIL-G-10924 For bearings where operating temperatures do not exceed 175 F. Bearings shall be cleaned thoroughly before lubrications because Military greases are not compatible with other greases. Each icemaking plant shall be tagged in a conspicuous place to indicate which grease has been used.

3.9 Fungus and moisture resistance.- The electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 0.001 inch to component or element surfaces previously cleaned and prepared.

* 3.10 Identification marking.- The icemaking plant shall be identified in accordance with MIL-STD-130. The marking shall be applied to the icemaking plant on plates conforming to MIL-P-514, type I, style 1, composition C. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

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* 3.11 Instruction plates.- Each icemaking plant shall be equipped with instruction plates, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the icemaking plant. Instruction plates shall conform to type III, composition A, class 1, of MIL-P-514. Plates shall be attached by screws, bolts, or rivets in a conspicuous location.

3.12 Treatment and painting.- The portions of the icemaking plant and its components and parts normally painted shall be cleaned, treated, and painted in accordance with type A of MIL-T-704, except as specified herein. Parts subject to extreme high temperatures shall be coated with a material conforming to MIL-P-14105.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certified test data.- When certified test data is submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for the defects listed in table I, for the dimensions specified herein, and tested, in order, as specified in 4.4.1 through 4.4.4. Failure to pass any test shall be cause for rejection of the preproduction sample. The presence of any defects listed in table I or any dimension not within specified requirements shall be cause for rejection of the preproduction sample.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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* 4.3.2 In-process inspection.-

* 4.3.2.1 Process examination.- Examination shall be made of the following processes to establish conformance with specified requirements. Whenever nonconformance is noted correction shall be made to the affected item and process.

(a) Dissimilar metals in contact (see 3.4.1).
(b) Pipe insulation (see 3.4.4).
(c) Brine tank finishing (see 3.5.7).
(d) Covers dipped in linseed oil (see 3.5.7.4).
(e) Cover construction (see 3.5.7.4).
(f) Treatment of electrical components (see 3.9).
(g) Grease provided (see 3.8).
(h) Can water tightness and zinc-coating provided (see 3.5.7.3).
(i) Seams welded smooth and coated (see 3.5.7.3).
(j) Insulation installed in accordance with 3.5.14.
(k) Determination of "U" factor (see 3.5.14).

$$U = \frac{1}{\frac{X_1}{K_1} + \frac{X_2}{K_2} + \frac{X_3}{K_3}}$$

Where U = overall heat transfer factor for the complete wall thickness.
X_1 = thickness of first material in inches.
X_2 = thickness of second material in inches.
X_3 = thickness of third material in inches.
K_1 = commercial K factor of first material.
K_2 = commercial K factor of second material.
K_3 = commercial K factor of third material.

* 4.3.2.2 Process tests.-

4.3.2.2.1 Insulation.- The compression test for the tank insulation shall be performed as specified in 4.4.5. The inspection lot shall be the material used in one days production. The sample unit shall be a 6-inch cube.

* 4.3.2.2.2 Moisture content.- The lumber moisture content shall be performed as specified in 4.4.6. The inspection lot shall be the lumber used in one days production. The sample unit shall be one piece of lumber. Three dimensions shall be taken on each sample unit and the results shall be the average of these determinations. The inspection level shall be S-1 with an acceptable quality level (AQL) of 2.5 defects per hundred units.

* 4.3.3 End item inspection.- The inspection lot shall be all ice making plants offered for inspection at one time. The sample unit shall be one completely fabricated ice making plant.

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* 4.3.3.1 Visual examination.- An examination shall be made of the ice making plants for defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Materials	Dissimilar metals in contact		X
	Not type specified		X
Design	Plant assembly not in accordance with drawing and instructions	X	
	Accessibility to components not provided	X	
Condenser	Component missing	X	
Receiver	Mounting supports missing		X
	Valve missing	X	
	Sight glass missing		X
Evaporator	Oil blow-off valve missing	X	
	Valve missing	X	
	Strainer missing		X
Agitator	Not propellor type		X
Cans	Seams not finished		X
	Not tapered		X
	Top edge not reinforced		X
Frame and covers	Defective wood		X
Motors	Motor rotations not shown		X
	Compressor motor less than 60 HP	X	
Controllers	Not enclosed type	X	
	Incorrect rating	X	
	Conversion equipment not provided	X	
Safety controls	Inadequate	X	
Electric hoist	Deficient capacity	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Can dump	Ineffective	X	
	Not automatic		X
Can filler	Not automatic		X
Ice chute	Defective flap		X
	Deficient capacity		X
Pipe	Type fittings		X
	Control valves missing		X
Cooling coil	Mounting hardware inadequate		X
	No hot-gas provision		X
Thermostat	Not provided as specified		X
Calcium chloride	Not provided as specified		X
Salt scale	Not provided as specified		X
Ice tongs	Not provided as specified		X
Thermometers	Not provided as specified		X
Refrigerant	Insufficient		X
	Cylinders inadequate		X
Welding rod	Insufficient		X
Platform scale	Not furnished		X
	Inadequate		X
Masks	Inadequate	X	
	Not furnished	X	
	Cannisters missing		X
Oil charging pump	Not furnished		X
Ice picks	Not furnished		X

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
pH tester	Not furnished		X
Lubrication	Not accessible		X
	Type fitting		X
	Type grease		X
Finish	Not color specified		X
Marking identification	Missing, incomplete or not as specified		X

* 4.3.3.2 Dimensional examination.- An examination shall be made of the icemaking plants for defects in dimensions and receiver volume. Any dimension not meeting specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

* 4.3.3.3 End item testing.- Each ice making plant shall be tested in order, as specified in 4.4.1 through 4.4.3. When no preproduction sample is required, the initial production unit shall be tested as specified in 4.4.4. If the manufacturer has previously manufactured and tested satisfactorily an identical ice making plant of this type and size for all requirements of this specification, in lieu of the performed testing of the completely assembled plant as specified in 4.4.3, the supplier may furnish certified test data showing that the plants he proposes to furnish are the same identical plants as the plant tested and that the requirements of the plant tested are identical to the requirements of this specification. Failure to pass any test shall be cause for rejection.

* 4.3.4 Code and standard compliance.- Proof of compliance with the requirements of 3.3.1, 3.5.3, 3.5.5, and 3.5.6 shall be submitted to the Government representative.

* 4.4 Tests.-

4.4.1 Refrigeration system leak test.-

4.4.1.1 Pressure test.- Prior to initial operation of the plant and in accordance with the manufacturer's instructions for valve operations, subject the entire system with an air pressure of 150 psi. Apply a strong soap and

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water solution to all flange and thread connections in the entire system; examine connections carefully for soap bubbles indicating leaks which constitute failure of the test.

4.4.1.2 Vacuum test.- After successful pressure testing, clean the system in accordance with manufacturer's instructions and subject the system to 25 to 30 inches of vacuum. Failure of the system to hold the vacuum for at least 6 hours constitutes failure of the test.

4.4.2 High pressure-low pressure cutout.- With pressure gage lines connected to the compressor service valves, operate the icemaking plant. Raise the condensing pressure by blocking off the condenser cooling. High pressure switch shall stop the compressor when the head pressure reaches 235 ± 10 psig. Reset the cutout control and restart the icemaking plant. Reduce the suction pressure by utilizing the pump down provisions to check operations of low pressure cutout. Cutout pressure should be zero pounds. Failure of the pressure cutouts to operate in accordance with the requirements as specified in 3.5.10 shall constitute failure of the test.

4.4.3 Performance test.- Operate the icemaking plant for a continuous period of 7 days at the conditions specified in 3.5. Determine the capacity for a period of not less than 100 hours. Any evidence of nonconformance to the requirements of 3.5, structual damage, misalinement or malfunctioning of components, refrigerant or brine leaks, erratic pressures, abnormal vibration, insufficient capacity, insufficient brine velocity (see 3.5.7.1), or irregular operation shall constitute failure of this test.

* 4.4.4 Electromagnetic interference.- The preproduction model or initial production unit when no preproduction model is required shall be tested for the reduction of radio interference in accordance with 3.7 and shall be tested by the supplier in accordance with Test Methods CE 03 and RE 02 of MIL-STD-462. The supplier shall furnish the test report to the contracting officer for approval and, provided all other requirements of the specification are met, the preproduction model shall be used as a model for all production icemaking plants.

4.4.5 Compression test for tank insulation.- To determine conformance with the requirements of 3.5.14, the tank-floor insulation shall be tested for compression strength in accordance with procedure A of ASTM Standard D1621-59T. Inability to meet compression strength shall constitute failure of test.

* 4.4.6 Moisture content.- Lumber shall be tested for the moisture content in accordance with ASTM D2016 to determine compliance with 3.5.7.4.

* 4.5 Examination of preparation for delivery.- An examination shall be made to determine that preparation for delivery as specified in section 5

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of this specification is in conformance with MIL-R-3593 and for defects specified in table II. The inspection level shall be S-2 with an AQL of 6.5, expressed in terms of defects per hundred units.

TABLE II.- Examination of preparation for delivery

Defects
Refrigerant compressor crankcase not drained and charged with a holding charge of dry nitrogen for level A.
Distribution panel and room thermostat not preserved as specified for level A.
Loose electrical wire not coiled and secured as specified for level A.
Ice cans not nested together and secured as specified for level A.
Insulation not packaged as specified for level A.
Electrical hoist motor not preserved as specified by the referenced document for level A.
Intermediate packaging not as specified for level A.

* 5. PREPARATION FOR DELIVERY

5.1 Preproduction pack.- A preproduction pack (Pilot Pack) shall be furnished as specified in MIL-R-3593.

5.2 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

5.2.1 Level A.- All uncoated ferrous metal surfaces and unassembled components of the icemaking plant shall be preserved and packaged in accordance with level A requirements of MIL-R-3593 with the following exceptions and additional requirements.

5.2.1.1 Refrigerant compressor.- The compressor crankcase shall be filled to operating level with the oil required for operation and the compressor operated to assure complete oil coverage of all working parts. The oil shall be drained and the compressor charged with a holding charge of dry nitrogen. The required quantity of oil shall be contained in the supplier's standard commercial containers.

5.2.1.2 Distribution panel and room thermostat.- The distribution panel and room thermostat shall be preserved together in accordance with method

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II b of MIL-P-116. The inner container shall be a box conforming to type CF, class domestic variety SW, grade as applicable of PPP-B-636. The outer container shall be a box conforming to domestic type, style optional of PPP-B-601. The outer container shall be lined with fiberboard conforming to type, class, and variety optional of PPP-F-320, to prevent damage to the barrier material. The lid of the outer container shall be secured with wood screws.

5.2.1.3 Loose wire.- Loose electrical wire shall be coiled to not less than 10 times its diameter, and the coils shall be secured with equally spaced wraps of cotton tape or twine.

5.2.1.4 Ice cans.- The ice cans required for each ice plant shall be nested together in quantities most convenient to accomplish the packing specified herein. The nested cans shall be secured together with flat steel strapping conforming to type I, class B, size as applicable of QQ-S-781.

5.2.1.5 Calcium chloride.- The calcium chloride shall be packaged in the supplier's standard commercial steel drum.

5.2.1.6 Insulation.- Insulation shall be packaged in close-fitting boxes conforming to style optional, W5c to PPP-B-636. The cementing material shall be packaged in the supplier's commercial container.

* 5.2.1.7 Electric hoist.- The electric hoist motor shall be preserved as specified for electric motors in MIL-R-3593.

5.2.1.8 Thermometers, salt scale, gas mask with extra cannisters, and miscellaneous operating accessories.- Items of like description shall be preserved together in accordance with method III of MIL-P-116.

5.2.1.9 Miscellaneous components.- Miscellaneous components not specifically specified in MIL-R-3593 or herein shall be preserved and packaged as specified for similar components.

5.2.1.10 Intermediate packaging.- Packaged and unpackaged components requiring the protection of a wood box to accomplish the packing specified herein shall be intermediately packaged in snug-fitting boxes conforming to domestic type, style optional of PPP-B-601 or class 1, style optional of PPP-B-621. The contents shall be blocked, braced, anchored, and cushioned as applicable within the boxes in accordance with MIL-STD-1186. Strapping of boxes shall not be required.

5.2.2 Level C.- The icemaking plant shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

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5.3 Packing.- Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A.- Each complete icemaking plant, preserved and packaged as specified in 5.2, shall be packed in accordance with the level A packing requirements for boxed equipment of MIL-R-3593.

5.3.2 Level C.- Each complete icemaking plant, preserved and packaged as specified in 5.2, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.4 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The icemaking plant equipment covered by this specification is intended for assembly and use in a theater of operations. Installation will be made in a standard 40-foot by 60-foot wood frame building. The Government will furnish electric power to the main switch and all wiring for lighting the building in which the plant will be installed.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) First article (see 3.1, 4.2 and 6.4).
(c) When isolation pads or springs are not required (see 3.5.1).
(d) Whether electrical characteristics are other than specified (see 3.5.8).
(e) Level of preservation and packaging and level of packing required (see 5.2 and 5.3).
(f) When the supplier is to furnish supervision of installation (see 6.6).

* 6.3 Drawings.-

6.3.1 Building.- The drawing, showing the construction and layout of the building, may be found in Department of the Army Technical Manual, TM5-302, Construction in the Theater of Operations, Drawing T 043-23, Storage Facilities Ice Plant, 15-Ton, 40' x 60' Wood Frame. This drawing may be obtained from the Chief of Engineers, Attn: ENGMC-EA Department of the Army, Washington, D. C. 20314.

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6.3.2 Iceplant.- The supplier shall be responsible for preparing drawings depicting the plant layout and erection drawings showing completely assembled components and equipment in place. The supplier shall submit to the contracting officer for approval the plant layout and erection drawings within the time frame specified. Approval of these drawings shall not relieve the supplier of his responsibility to furnish equipment conforming to this specification. The drawings shall depict but not necessarily be limited to the following:

(a) Bill of materials.
(b) Master flow diagram.

(1) All valves, gages, controls, etc., shall be identified with their symbols or nomenclature.
(2) Direction of flow shall be indicated.
(3) Type of flow (high, low pressure), liquid or gas, water, brine, oil, refrigerant, etc.

(c) Plan view.

(1) Location of components.
(2) Mounting requirements for components.
(3) Dimensions, length, and size of pipes, valves, insulation, etc.

(d) Electrical schematic.

(1) Controls.
(2) Valves.
(3) Wiring diagrams.

Two complete sets of plant layout and erection drawings shall be furnished with each icemaking plant.

* 6.4 First article.- The item covered by this specification requires First Article test and approval under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The First Article should be a preproduction sample. The First Article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examination, test and approval of the First Article.

6.5 Contract data requirements.- Any requirements for equipment manuals for the item covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

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6.6 Installation supervision.- When specified (see 6.2), supervision of installation shall be supplied by the contractor for ice making plants to be erected in the theatre of operations. The contractor shall furnish the services of a competent superintendent who should be familiar with the construction, erection, operation maintenance, and repair of the equipment and the required utility connections. Material required for complete installation, other than that specified herein, and all installation labor will be furnished by others. All utilities will be extended and connected to the equipment by others. The superintendent shall supervise the installation of the equipment and all utility connections thereto, using the material furnished as stated. Supervision of utility connections shall include all work from points within 10 feet of the ice making plant. Upon completion of the work, the superintendent shall turn the equipment over to the contracting officer or his representative, in proper operating condition, and shall instruct personnel designated by the contracting officer in the proper operation.

6.7 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to that last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 82

Review Activity:

Army - MD

User activities:

Army - CE, ME

Preparing activity:

Army - GL

Project No. 4110-0211

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Icemaking Plant: Block MIL-I-10493D

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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